

Tissue organisation, communication, and interpretation

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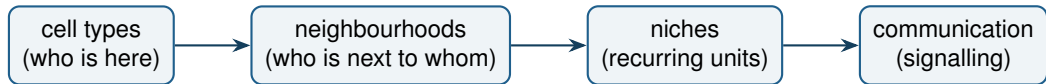
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Day 3 – 23.06.2026

Today, and why it stands alone

- We run on a separate, smaller machine (6 GB) and a small, pre-annotated single-cell spatial dataset. Nothing today depends on Days 1–2.
- Focus shifts from *making* the data to *reading* it: organisation, communication, and careful interpretation.
- These are the analyses that turn a cell-type map into biology.

From cell types to tissue organisation



Each step is built from the spatial graph. Proportions alone miss the arrangement, which is where tissue function lives.

Neighbourhood enrichment and co-occurrence

- **Enrichment:** permute cell-type labels on the graph; compare observed adjacency to the null. High means co-location, low means avoidance.
- **Co-occurrence:** how the chance of finding type B near type A changes with radius, separating contact from regional association.

Caveat: both are relative to the chosen graph and the section's composition, so a difference between samples can be a composition difference.

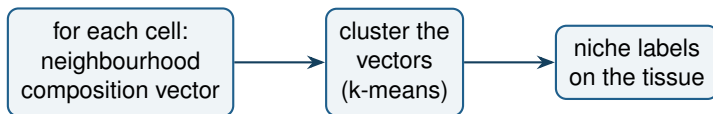
Niches and tissue domains: the method landscape

- **Composition clustering** (what we do): summarise neighbourhood cell-type composition, then cluster. Transparent, fast, CPU-friendly.
- **BANKSY**: augments each cell with its neighbourhood; types cells and finds domains together, scalable and batch-aware.
- **CellCharter**: clusters neighbourhood-aware embeddings into niches across samples.
- **SpaGCN, STAGATE**: graph neural networks for spatial domains.

Learned methods can capture subtler domains; they add assumptions and compute.

Singhal et al., *Nat Genet* 2024 (BANKSY); Varrone et al., *Nat Genet* 2024 (CellCharter); Hu et al., *Nat Methods* 2021 (SpaGCN); Dong & Zhang, *Nat Commun* 2022 (STAGATE).

Building niches transparently

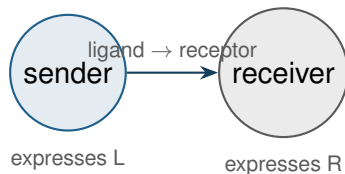


The output is named, mappable units you can compare across samples, with no black box between the data and the label.

Cell–cell communication: the logic

Score ligand–receptor pairs between cell-type groups: is the ligand expressed by one type and the receptor by a co-located type?

In spatial data we can add the constraint that the partners are actually neighbours, which dissociated data cannot.



Communication: methods and caveats

- **Non-spatial:** CellPhoneDB, CellChat, NicheNet (adds downstream targets); LIANA unifies many under one interface.
- **Spatial-aware:** squidpy ligrec, stLearn, COMMOT (optimal transport), NCEM (niche effects on expression).

Caveats: expression is a proxy for activity; a small panel covers few pairs; p-values depend on the permutation scheme; segmentation errors move ligands to the wrong cell. Read results as hypotheses.

Efremova et al., *Nat Protoc* 2020 (CellPhoneDB); Jin et al., *Nat Commun* 2021 (CellChat); Browaeys et al., *Nat Methods* 2020 (NicheNet); Dimitrov et al., *Nat Commun* 2022 (LIANA); Cang & Nie, *Nat Methods* 2023 (COMMOT); Fischer et al. 2023 (NCEM).

Spatially variable genes and pattern reading

- Moran's I ranks genes by how smoothly they vary across the tissue.
- Top genes often trace anatomy or gradients rather than discrete clusters, which complements clustering.
- Plot the top hits in space and ask what structure each marks.

Comparing conditions

- With a disease cohort, the question becomes which niches or compositions change between groups.
- Composition-aware tests (e.g. Milo for differential abundance, scCODA for composition) avoid the pitfalls of comparing raw proportions.
- Spatial features (niche burden, distances, enrichment scores) can be summarised per sample and tested like any covariate.

Dann et al., *Nat Biotechnol* 2022 (Milo); Büttner et al., *Nat Commun* 2021 (scCODA).

- Segmentation errors propagate into co-expression and communication.
- Low cytokine and chemokine detection makes immune states uncertain; absence is weak evidence.
- Spatial QC metrics are still maturing; thresholds are provisional.
- Composition differences between sections can masquerade as enrichment. Compare like with like.

- Integrating large H&E images with the molecular layer, and pathology image foundation models for region identity (e.g. UNI, Virchow, H-optimus), with attention to their failure modes.
- Multimodal imaging plus Xenium to give better identity to regions of interest.
- Agentic and LLM-assisted analysis workflows.

These are open and moving quickly; today's tools are a foundation, not the ceiling.

Further reading

- Varrone et al. CellCharter. *Nat Genet* 2024; Singhal et al. BANKSY. *Nat Genet* 2024.
- Cang & Nie. COMMOT. *Nat Methods* 2023; Dimitrov et al. LIANA. *Nat Commun* 2022.
- Dann et al. Milo. *Nat Biotechnol* 2022.
- Squidpy tutorials (neighbourhood enrichment, co-occurrence, ligrec).

Dataset: a small pre-annotated seqFISH section. You will:

- run neighbourhood enrichment and co-occurrence;
- identify niches from neighbourhood composition;
- score ligand–receptor interactions (offline);
- rank spatially variable genes.

Capstone: pick one finding and read it end to end – what it claims, whether it is consistent, what could confound it, and what would test it next. We discuss as a group.